UNIT 2 INTRODUCTION TO MOBILE COMPUTING ARCHITECTURE

2.0 INTRODUCTION

Nowadays, mobile computing is one of the fastest-growing technologies. A major strength of this service is its broad reach and ease of use. In addition to the computing requirements and new apps in mobile devices, the number of mobile device users is also growing exponentially, as is the demand for resources for them. As a result, the design and construction of a mobile computing architecture that is efficient, effective, scalable, and secure is one of the most challenging aspects of the mobile computing environment.

2.1 OBJECTIVES

At the end of the Unit, you shall be able to:

1. Understand various types of mobile computing networks that exist.
2. Understand the development of novel applications, the use of smart phones and the Internet, enterprise solutions, mobile personal cloud, mobile payments, and mobile wallets.
3. Understand Three-tier architecture and N-tier architecture for mobile computing.
4. Understand design considerations in mobile computing including operating systems, languages, protocols, software layers, and data synchronization and dissemination.
5. Understand how to incorporate mobile computing into existing applications through the Internet.
6. Understand various limitations of Mobile devices.
7. Understand Mobile computing security.

2.2 MOBILE IP, CELLULAR AND WLAN WI-FI IEEE 802.11x NETWORKS

Mobile IP: A request for comments (RFC) 2002 was issued by the Internet Engineering Task Force (IETF) defining mobile IP as an open standard. All media-supporting IPs also support mobile IP due to it being based on the Internet protocol (IP). Agents at home and abroad provide the mobile IP service through mobile IP networks. Distributed computing and mobile devices form mobile networks.

2.2.1 Cellular Network

There is a base station for each cell. Mobile devices use base stations as access points. A base station's coverage area defines a cell. Each cell has a defined coverage area. Wireless communication takes place between a mobile device and a base station within the cell boundaries. Each cell in a cellular network has an interconnected base station.

Every mobile service region consists of a number of cells. Based on the technology and frequency bands used within a cell, the size of a cell varies. As an example, the cell radius in CDMA 950 MHz networks is 27 km, while it is 14 km in CDMA 1800 MHz networks.

Assuming the cells are hexagonal in shape; cell A0 is surrounded by the boundaries of 6 cells: A1, A2, A3, A4, A5 and A6.

A0, A1, A2, A3, A4, A5 and A6 have base stations BS, BS1, BS2, BS3, BS4, BS5 and BS6, respectively. An area of coverage defined by A's cell boundaries is covered by BS.
An ith base station, BSi, can be considered as an access point for the services in the region Ai. Cells are centered on the base station. The base station is the only point of communication between mobile devices within a cell.

A guided network (wired or fibre-based) or wireless network connects the base stations. Alternatively, stations can connect to the public switched telephone network (PSTN). Switching is the act of establishing a connection and maintaining it until it is disconnected or switched off. There are a number of public services that provide the PSTN to a mobile service provider. Public telephone networks are extensive in the public service.

Transceivers (mobile phones) on a multi-cell cellular network must switch between cells when they move from one place to another. An initial cell's base station hands over a mobile device when it reaches a cell boundary. In order to switch to the next cell, the device connection is handed over to the neighboring base station.

Depending on the cellular network type (GSM or 3G CDMA), different handover mechanisms are used. Mobile devices switch from one channel to another without disrupting ongoing communication.

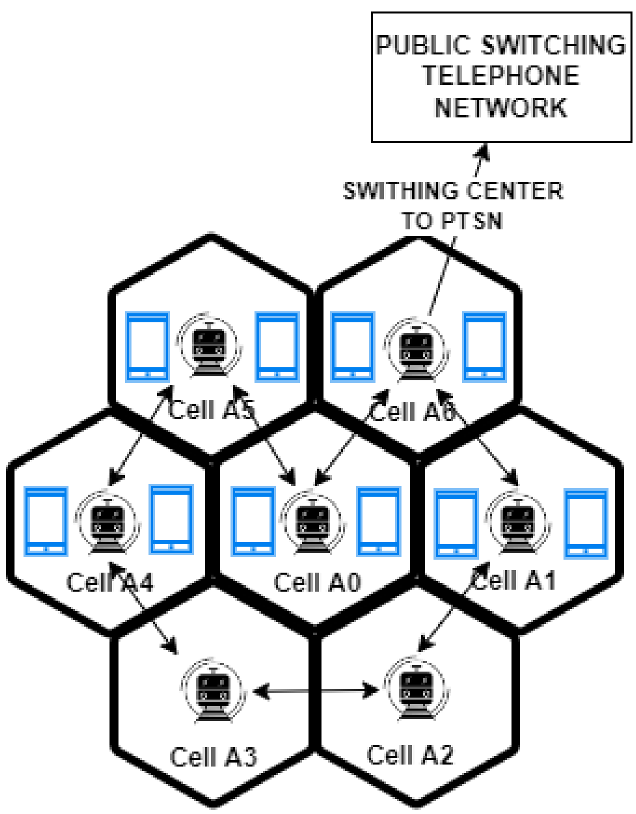

Fig 1: Mobile communication using the cells A1, A2, A3, A4, A5, and A6 of a cellular network

Frequency Reuse and Adjacent Channel Network

GSM communication technology uses different frequency bands for neighboring cells. Cells adjacent to the immediate neighboring cell can reuse the frequencies.

Throughout a cellular network, cells surround each other. Interference is caused when the same frequency band is used at the same time and at the same point. A cell's frequency differs from that of its neighbor. In FDMA, cells that reuse the same frequency channel must have at least one gap between them. It is necessary to use different frequencies for cells A1, A2, A3, A4, A5 and A6 if cell A0 uses frequency f0. By doing so, different cell signals will not interfere with each other. Whenever two frequencies are equal or integral multiples of one another, interference occurs.

Consider the cell A0 uses frequency f0 and cell A1 uses f1, cell A2 uses f2 then cell A3 can reuse f1. This is because; A1 and A3 do not lie adjacent to each other, so there is a one-cell gap between them. Similarly cell A4 can reuse f2, cell A5 can reuse f1 and cell A6 can reuse f2. Whenever frequencies are reused, they need to be allocated separately, like f0, f1, and f2. There is a 1/3 frequency reuse factor. Keeping at least one cell separation allows frequency reuse in this case. A cell gap therefore exists between cells that reuse the same frequency channel.

It is possible to have a frequency reuse factor (u) of 1/3, 1/4, 1/7, 1/9, or 1/12. Using more frequencies in the cells can also reduce cell sizes.
Frequency reuse formula: distance $d = r \sqrt{(3 \times n)}$
In this equation, r refers to the distance between the center of the cell and the boundary, and n refers to the number of cells surrounding the cell.

From the perspective of the base station (BS), each cell can be divided into sectors. Suppose that a BS uses m antennae per sector. Different antennas can use the same frequency and point in different directions. During space division multiplexing, the cell divides its space. As a result, the frequency reuse factor will be m/u. Reuse patterns of 3/4 are used in GSM mobile networks. Consider a GSM service with a total bandwidth of b. Therefore, b/u is the number of frequency channels available. In space division multiple access (SDMA), each sector can use bandwidth b, which is equal to b/m x u.

A set of frequency channels can, therefore, be used by GSM mobile service networks. Cells adjacent to each other are allocated different frequency channels.

A CDMA network uses a spread spectrum (SS), which means that a range of frequencies can be used; however, they are used in conjunction with a coding scheme. For example a school proposes three colors of dresses and each class uses these three colors. Thus, the scheme can be: class 1 has blue on Mondays, black on Tuesdays and purple on Wednesdays. For class 2, it can be black on Mondays, purple on Tuesdays, blue on Wednesdays and so on. Each user's channel, or frequency in the spectrum, is coded differently, so all frequencies are available, but with a different coding scheme.

A frequency reuse factor of 1 is applied for each sector and each cell when the same spectrum is used, but the code used to encode the chipping frequencies or frequency hopping sequences is different.

There is a high data transfer rate between pre-4G and 4G services. Using multiple antennas that share bandwidth, multiple antennas communicate coherently with the user's mobile device. Distance and path length increase incoherency. Pico cells reinforce the coherency and relay it to the devices when each cell sector is divided into picocells. As a result, pre-4G services divide networks into narrow regions by using picocells, and further divide smaller networks by using femtocells. It is possible that one picocell corresponds to one floor of a building.

Capacity Enhancement in Networks

Frequency reuse enhances capacity. Additionally, multiplexing increases capacity. It is possible to share data transmission space, time, frequency, or code between different channels, users, or sources. In terms of space, time, and frequency, multiplexing (SDMA, TDMA, and FDMA) determines how these resources can be shared.

Enhancement of capacity caused by frequency reuse is given by k x m x u. In this equation, k is the enhancement due to multiplexing, m is the number of sectors within a cell, and u is the number of adjacent cells.

The beams transmitted by a sector antenna can be divided into micro sectors. Signals radiated from the antenna are referred to as beams. Further capacity can be enhanced by using switched beam smart antennae. There are p different directions in which an antenna can radiate a beam. Frequency reuse increases capacity by p x k x m x u.

Co-Channel and Adjacent Channel Interference

Two signals of the same frequency or very close frequency superimpose at an instance at a place when they use two channels, and when the phase difference between both signals is 180 (or odd multiples of ∏), then the amplitudes of the signals subtract, resulting in 0. In a case where the phase difference is 0° (= even multiples of ∏), the amplitude adds up and the resultant amplitude is twice the amplitude of the individual signals.

It is possible for sources to use several channels at the same time. Signals with close frequencies are susceptible to interference. The phenomenon is known as co-channel interference (cross-talk). The interference effects between two channels with P1 and P2 are negligible when P2 <<P1.

Let's look at an example. Interference effects between two people speaking near each other in a feeble voice and one in a louder voice are negligible. Re-using frequencies and high co-channel power levels increase interference between co-channels.

Signals from nearby sources cause adjacent channel interference.

Two modulated sources transmitting in the same band overlap their frequencies, causing narrow band interference. Additionally, interference between adjacent channels can be caused by insufficient frequency control or tuning in a source.

As the individual spectral carriers have reduced power, spread spectrum reduces co-channel interference. As a result of the spread spectrum's frequency bands, power is distributed over a greater number of frequencies. Due to the large number of frequencies used in narrow band interference, it has too little effect.

Cellular Broadband

There are now 3G-enabled cell towers that support EV-DO, HSDPA, and HSUPA. Therefore, mobile broadband access is supported by the service providers. It is possible to accomplish this with the help of a USB cellular modem or a cellular broadband router. Multiple computers can be connected by a router. A USB cellular modem connects to a computer. As with a pen drive, the modem is attached via a USB port.

Mobility Management

In order to maintain continuous (seamless) signal connectivity when a mobile device moves from a cell, Ci, or network Ni, to a cell, C1, or network N1, mobility management is necessary. The following must be present to guarantee ongoing connectivity:

1. Management of the infrastructure for setting up and maintaining the links between networks Ni and N1 or cells Ci and C1.
2. When a mobile device's connection with the i^{th} cell is transferred (on handoff from the i^{th} cell) and registered at the new (j^{th}) cell, location and registration management by handoff for cell transfer is used.

2.2.2 WLAN Wi-Fi IEEE 802.11x Networks

Wireless local area network (WLAN) sets of well-liked standards have been suggested for mobile communication. The standards mentioned here are IEEE 802.11a, 802.11b, 802.11g, 802.11i, and 802.11n. When mobile devices, iPads, laptops, desktop computers, or printers connect to an access point utilizing a protocol standard outlined in IEEE 802.11x, where x = a, b, g, I or n, a wireless LAN has been created. WLAN is a Wi-Fi-based wireless network service.

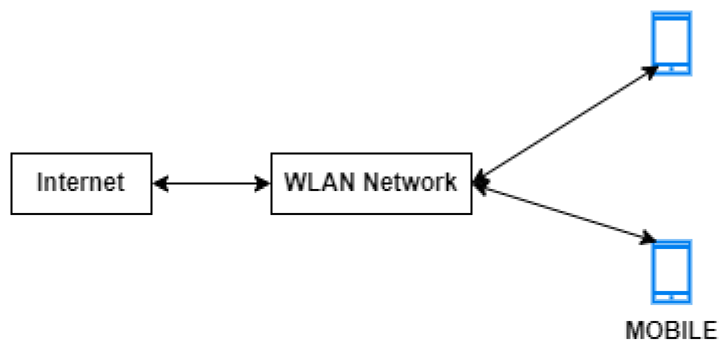

Fig2: 802.11 WLAN communications

<u>WLAN and Internet Access</u>

Communication between mobile devices and the Internet is established by the network. Mobile devices and the Internet can communicate. Thanks to the network.

1. An iPad, tablet, or other mobile device connects to the Internet by way of an Internet service provider using Wi-Fi. A mobile device, such as a tablet, iPad, or laptop, connects to a hotspot-style access point. A router connects the host LAN, which is connected to the Internet by the access point, to the Internet. As a result, connectivity develops throughout the Internet. Computers, mobile devices, and two LANs.
2. Web content is sent to mobile phones' small-area display devices using the wireless application protocol (WAP). The WAP format is used by the service providers to format material.
3. The screens of modern mobile devices, tablets, and iPads are larger. Additionally, the majority of modern devices can connect to the Internet via HTTP using either a mobile data service provider or an Internet service provider. Many modern devices support both HTTP and WAP. For instance, NTT DoCoMo in Japan developed Internet in Mobile Mode (i-Mode), which was a hugely well-liked wireless Internet service for mobile phones.

Standard	Extension	Description
802.11	a	There may be more than one physical layer in 5 GHz due to MAC layer operations (infrared, two 2.4 GHz physical layers). The layers enable both mobile ad hoc network (MANET)-based architecture and infra structure-based architecture. At data rates of 6 Mbps and 9 Mbps, OFDM modulation is used. The supported data rates range from 54 kbps to a few Mbps.
	b	It uses DSSS/FHSS modulation at 2.4 GHz and works at 54 Mbps. Additionally, it supports Bluetooth (IEEE 802.15.1)-based software and the HIPERLAN2 (HIPERformance LAN 2) OFDMA physical layer for use with short-range wireless networks. It offers secure Wi-Fi connection. 1 Mbps (Bluetooth), 2 Mbps, 5.5 Mbps, 11 Mbps, and 54 Mbps are the available data rates (HIPERLAN 2)
	g	It runs at 2.4 GHz and 54 Mbps. It is compatible with 802.11b and used for many new Bluetooth applications. OFDMA is replaced by DSSS.
	i	The AES and DES security standards are provided.
	n	A new multi-streaming modulation method is IEEE 802.11n. Numerous new capabilities are included, including multiple-input multiple-output (MIMO) antennae. Using several 54 Mbps streams, data rates can be increased to 600 Mbps. It allows the usage of four spatial streams with a carrier frequency of 2400 MHz and a channel width of 40 MHz. It supports 64 QAM, QPSK, and BPSK.

Table 1

2.3 AD HOC NETWORKS

A temporary variety of local area network is an ad hoc network (LAN). An ad hoc network turns into a LAN when it is permanently installed. An ad hoc network may accommodate multiple users at once, although performance may suffer. As long as the hosting device has internet connectivity, users can also use an ad hoc network to connect to the internet. This could be helpful if numerous individuals need to access the internet but there is only a limited amount of internet connectivity in a certain location.

Through an ad hoc network, multiple devices can share the host device's internet connectivity. Jobs that handle this kind of network are often well-paid by employers, especially in professions that require a lot of travel.

Security is one of the main issues with an ad hoc network. Cybercriminals can typically connect to a wireless ad hoc network and, consequently, to the device if they come within signal range.

The network name cannot be disguised if users are in a public area since users cannot prevent their SSID broadcast in ad hoc mode. Ad hoc networks are not always appropriate because of this.

Ad hoc connections, however transitory and only reachable within 100 metres, can be useful in some circumstances. Attackers are unable to access a gadget from a distance and are limited in the amount of time they have to plot their attack.

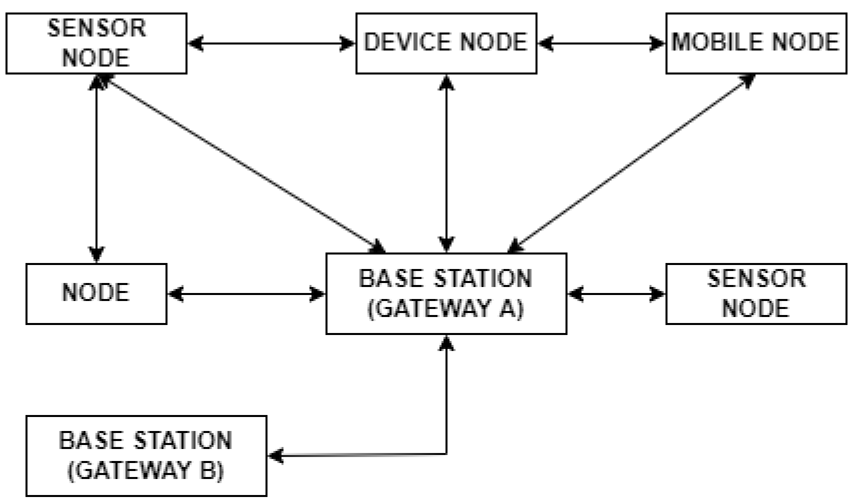

Fig3: Direct communication between sensor nodes and mobile nodes utilizing a base station as a gateway

The above diagram demonstrates how a base station is used to facilitate communication between the nodes, device node, mobile nodes, and sensor nodes. Gateways are acted upon by the base stations.

Each node has the ability to discover itself and configure itself. Every node has a router capability. A router locates the additional communication channels that are available. Ad hoc networks are used in a mobile context for routing, target detection, service discovery, and other requirements.

2.4 MOBILE COMPUTING OPERATING SYSTEM

An operating system is necessary for mobile computing. Without taking into account the capabilities and specifications of the hardware, an operating system (OS) enables the user to run an application. Additionally, it offers tools for organizing various tasks into a system's schedule. Even the personal information manager (PIM) and APIs for using SMS, MMS, GPS, and other apps are provided by an OS.

Tasks and memory can be managed by an OS using tools including creation, activation, deletion, suspension, and delay. It offers the features necessary for the system's many tasks to be synchronized. There may be several threads for a task. It offers thread synchronization and priority distribution.

Additionally, an OS offers interface for software at the application layer, middleware layers, and hardware devices to communicate with one another. It makes it easier for software components to run on a variety of hardware. For the device's graphic user interface (GUI), an OS offers programmable libraries. Many user-operated devices require User application's GUIs, Voice User Interface (VUI) components, and phone Application Programming Interface (API).

Device drivers for USB, keyboard, displays, and other devices are also provided by an OS. Middleware, applications, and a fresh environment for developing applications are all provided by mobile OS.

EXERCISE:

Question 1: What are cells and base stations?

Question 2: Describe how the cellular network operates. How a network's capacity is increased using the provided set of frequencies?

Question 3: What are the differences in characteristics provided by Bluetooth, ZigBee, and IrDA?

Question 4: Give an explanation of the wireless personal area network protocols (WPAN). Offer some WPAN network applications for the house.

Question 5: When is NFC used? Describe the uses for NFC.

2.5 CLIENT SERVER COMPUTING USING MOBILE

Think of a network of nodes that are spread out (computers and computing devices). A node may be either a client or a server depending on the network architecture. Application software is run on a client node and is dependent on server node resources (files, databases, Web pages, other resources, processor power, or other devices or computers connected or networked to it). The server node has more computer power and resources than the client nodes. The client-server computing architecture is the name given to this design. It differs from peer-to-peer design, where each network node has a similar set of resources and the different nodes rely on one another for those resources.

A distributed computing architecture known as client-server computing uses servers and clients as its two types of nodes. The server can be asked by a client for information or responses, which the client can subsequently employ in calculations. The client can cache these records on the client device or access them directly from the server. The data may be accessible at the client's request, via broadcasts, or by server distribution.

Due to their limited resource availability, mobile devices operate as client nodes. Several devices are connected to a server. Both the client and the server may be running on the same computer system or on distinct ones. An N-tier design for client-server computing (N = 1, 2) is possible. The number of tiers, N = 1, when the client and server are on the same computing system (not on a network). N=2 is the case when the client and the server are on different computing platforms connected to the network. N > 2: if the server is connected to or connected through networks to other computing systems that supplies the server with additional resources for the client. N> 1 denotes a connection between a client device at tier 1 and a server at tier 2, which may then connect to tiers 3, 4, and so forth.

For data, the client device synchronizes or connects to higher levels. To obtain client requests at the server or server responses at the client, a command interchange protocol (like HTTP) is used. The following list describes 2, 3, or N-tier architectures for client-server computing. A connecting, synchronizing, data, or command interchange protocol is used to link each layer to the others. The Java Remote Method Invocation (RMI) protocol and the C++-based remote procedure call are two examples of data or command transfer protocols (RPC).

Client-Server Two-tier Architecture

An application server may be required by a particular application to distribute a local copy of data to numerous devices. On the devices, the application can now execute independently without the need for run-time retrieval.

Through a synchronization API, the multiple APIs communicate with one another. When records are amended at the server end, synchronization means that the cached copies on the client devices should also be updated. Since different devices may use different platforms, the APIs are created as independently of hardware and software platforms as possible.

Three-tier Architecture for Mobile Computing

In three-tier computer architecture, the application interface, functional logic, and database are maintained at three distinct tiers. Through a synchronization-cum-application server at tier 2, data records at tier 3 are transmitted to layer 1 of the data chain. With the help of business logic, the synchronization-cum application server's synchronization and server programs retrieve data records from the enterprise layer (tier 3). Using a connectivity protocol, the enterprise tier connects to the databases and transfers the database records to tier 2 in accordance with the business logic query. Additionally, a three-tiered data hoarding system is displayed. A middle server known as the synchronization server transmits and synchronizes copies to numerous mobile devices. For the applications 1 to j, local copies of the databases 1 to I are stored on the mobile devices. In the image, an enterprise database connection to a synchronization server is also shown. This server synchronizes the enterprise database server with the local copies needed by the apps. In the event of a Web-client at tier 1, HTTP can also be utilized as a synchronization protocol.
Internet-based communication from Tier 1 to Tier 2 allows for further connectivity to be created using HTTP or HTTPS.

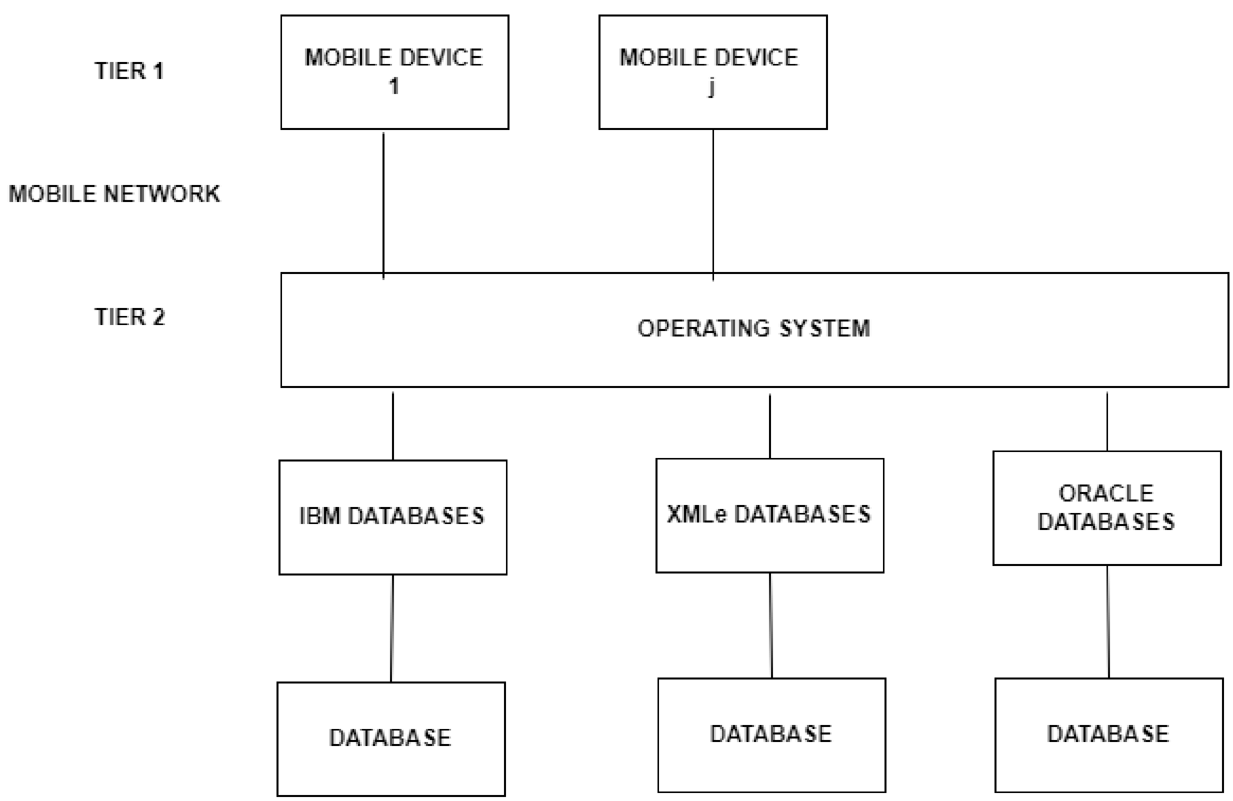

Fig 4: Two-tier client server application

2.6 MOBILE COMPUTING ARCHITECTURE

The architectural layers of mobile computing devices, mobile computing on the device using APIs are all described in this part of the Unit.

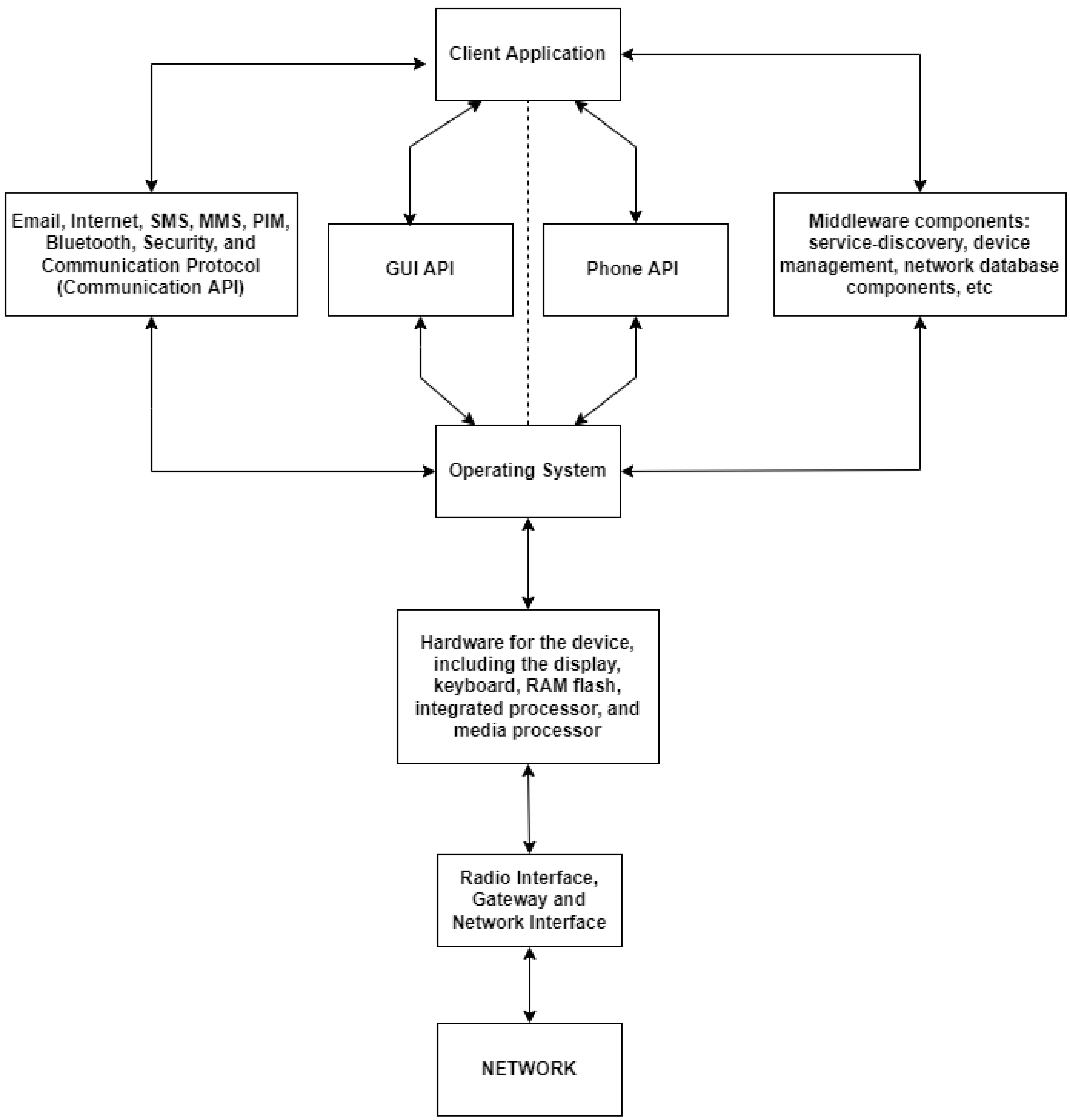

Fig 5: Mobile Computing Architecture

2.6.1 Design considerations for Mobile Computing

The architecture of mobile computing relates to the several layers that are established between user applications, interfaces, devices, and network hardware. For systematic calculations and access to data and software objects in the layers, a clearly defined architecture is required. The software components and APIs are deployed by an application. For instance, the communication APIs in a mobile smartphone's Internet, PIM, SMS, MMS, Bluetooth stack, security, and communication protocol stack. The client APls can be viewed as one of the architectural layers. In addition to updating databases, managing devices with remote server software, performing client-server synchronization, and adapting applications to certain platforms and servers, middleware components also locate services and connect client and network services. Between the program and the hardware lies a layer called an OS. It makes running the software easier, hides hardware characteristics, and offers numerous OS functionalities (e.g., device drivers). A program can directly use an OS function, for instance, the time delay (3000) function, which delays the execution of subsequent instructions by 3000 clock ticks.

2.6.2 Mobile Computing using APIs

The following are a few examples of mobile computing APIs: voice recognition, text-to-speech conversion, voice-based dialing, camera, album, video clip recorder, and Wi-Fi network access point connectivity.

EXERCISE:

Question 1: A client-server computing architecture with the database at the application tier should be displayed. When the application server retrieves the data from the enterprise server layer, how does this architecture change?

Question 2: Explain the four-tier architecture. In client-server architecture, how are multimedia databases used to serve a mobile device?

Question 3: What software components and layers are required in a mobile computing device?

Question 4: Explain the functionality of an OS in a mobile system.

Question 5: Describe a unique mobile application that interests you. What further software components are included in this application?

2.7 DESIGN CONSIDERATIONS FOR MOBILE COMPUTING

The architectural specifications for programming a mobile device are described in this section. It gives an overview of the programming languages used to create the software for mobile devices. To run the software components on the hardware, an OS is necessary.

1. **Considerations for Frameworks and Programming Languages:** The mobile computing architecture employs a number of programming languages. Java is one of the most used languages for mobile computing. This is due to Java's most significant feature, platform independence, which means that Java program codes are independent of the CPU and OS utilized in a system. This results from a typical compilation into bytes. J2SE is the name of Java 2's standard edition. There are two editions with a small memory footprint: Java2 Micro edition (J2ME) and Java Card (Java for smart card). These are the two most popular languages for mobile computing and creating apps for platforms used by mobile devices. Mobile service apps that are Web- and enterprise server-based employ Java 2 Enterprise Edition (J2EE).
   The other two widely used programming languages are C# and JavaScript. Program compilation for these languages depends on the CPU and operating system being utilized. The benefit of C# is that it makes it possible to build Uls, and JavaScript makes it possible to create contents on the fly.
   Server communications and client messages are presented using the markup language XML HTML5.
   Popular mobile application frameworks like Python 2.7 and 3.0 support scripting languages like JavaScript as well as programming languages like Java or C#.
   A framework for creating Internet and Web applications is provided by QtScript. For enterprise applications, J2EE is used, and for web pages, JSP.
   DotNet offers a platform for creating apps, web applications, and Internet services. For enterprise applications, it uses Visual Studio and ASP.NET, while for web pages, it uses JSP.

2. **Operating System Considerations:** When designing an app, an OS that is compatible with the mobile device is taken into account. The user can launch an application on the iPhone 6. Thanks to iOS 8, which is used by an app designer for an iPhone app. The following are some examples of how an app creator makes advantage of the OS's functions:
   a) Hardware features and specs
   b) Supporting the creation, activation, deletion, suspension, and delay of threads and the scheduling of numerous threads in a system
   c) Interfaces for exchanging information between hardware devices, middleware layers, and application layer software components

d) Facilitating the hardware's ability to run software components
e) Adjusting the device's library settings for the GUI and VUI components
f) Artistic writing, music composition, and graphic design (for example, OS X)

3. **Middleware functions:** Middleware is a term used to describe the computer programmes that connect application components to network-distributed components. Between a user application and an operating system, middleware functions as additional software. Additionally, Mobile OS offers middleware components. The following uses are made of middleware applications:
a) In order to find adjacent Bluetooth devices
b) To find the hotspot nearby
c) To synchronize a device with a server or an enterprise server
d) To retrieve data from a network database, which may be in Oracle or DB2 format
e) For transaction processing
f) To use an application server for a stateless Internet session
g) For service discovery
h) For application platform and service availability adaption

4. **Data Synchronization and Dissemination:** A mobile phone can be used as a data access device to access the server of the service provider and retrieve information. Smartphones serve as enterprise data access devices in networks used by businesses. The data is distributed to the enterprise mobile device, such as the BlackBerry handset, by an enterprise server.
A data access device for accessing music or videos is something like an iPod, iPhone, iPad, or tablet. Files can be downloaded via a link and saved or played afterward. These days, students can record lectures from professors and access e-learning materials using an iPod, iPhone, iPad, or tablet. To communicate, disseminate, or broadcast information, a data dissemination service is necessary. An example of a distribution server for academic lectures, interviews, and learning materials is https://itunes.standford.edu. The same service offers a music, iTunes, and video user interface as well. Media played on iPods is stored on a platform called iTunes, which is a program that can be downloaded from the Internet. At www.apple.com/itunes, Apple distributes music and video content for iPads and iPhones. Application servers, business servers, iTunes servers, and service providers' servers all send data to mobile devices. The three ways for disseminating data are: (i) broadcasting or pushing (such as sending unwanted SMSs to mobile phones), (ii) pulling (such as downloading a ringtone from the mobile service provider), and (iii) a push-pull hybrid.
Podcasting is a modern type of broadcasting. It is a brand-new technique for sharing multimedia files. For instance, files for music videos and audio programs are disseminated online in formats that can be played on computers and mobile devices. Receiving the multicast or unicast data that is distributed over a mobile network requires

middleware. Additionally, middleware offers data management service transparency and application adaptation. The user does not have to configure the underlying protocols thanks to transparency.

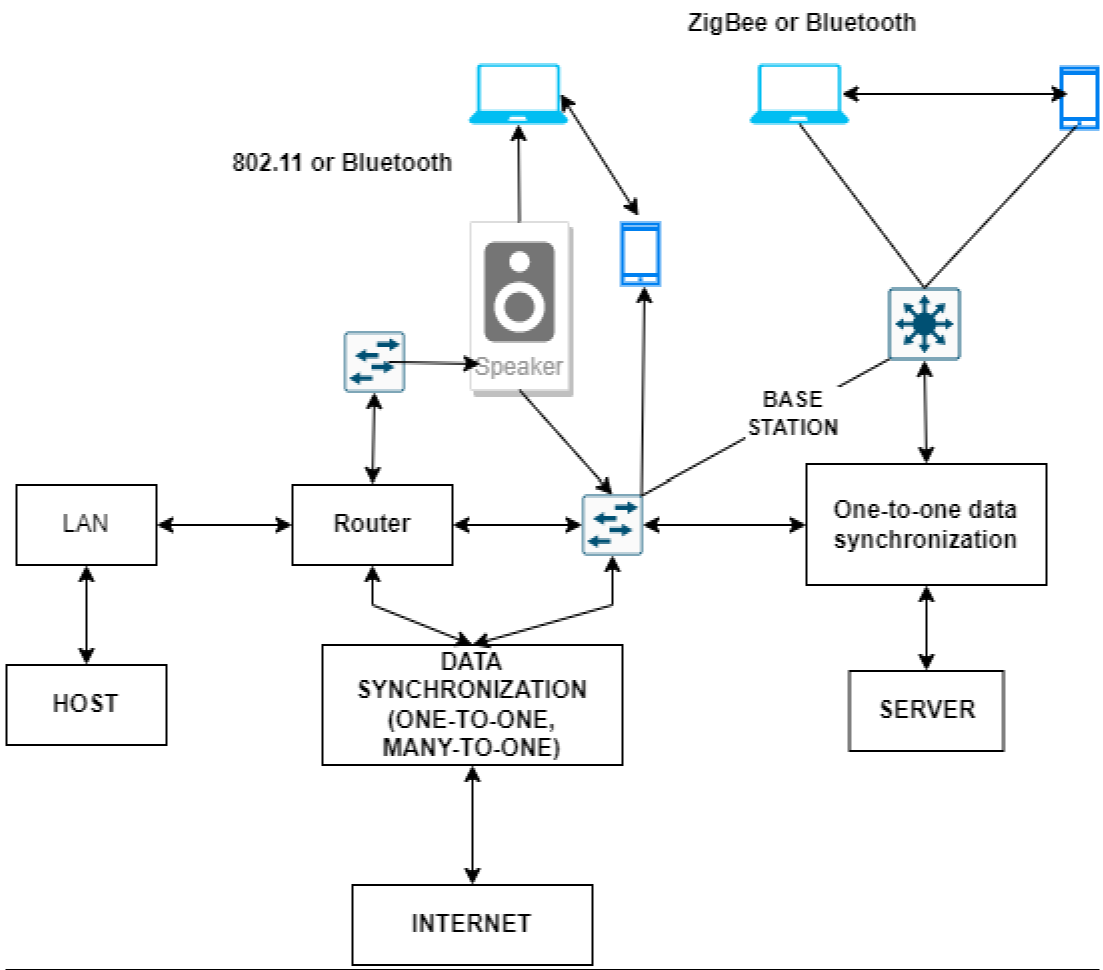

Fig 6: Data synchronization paths

<u>Synchronization</u>

Data synchronization, often known as synchronization, is the process of updating data across many databases so that each repository includes the same data. Data synchronization ensures that all servers of a mobile service provider have the same selection of ringtones if, for instance, a new, well-liked ringtone is added to one of the servers of the service provider. Furthermore, any new data should be made available to all the devices linked to the server. This indicates that a duplicate of the tone's title can be found in the ringtone database that is accessible to all mobile phones.

One-to-one synchronization: When two ends of a data stream are in sync, every modification made at one end is immediately reflected at the other. The information on the two ends should be the same or consistent. In a data-synchronized enterprise network, the change is reflected at the enterprise servers, for instance, if the address of a mobile device user changes and the user enters the new address in the device's address book. In this situation, a GSM or CDMA network is used to synchronize devices. But when an iPod is put next to a PC and iTunes is downloaded into the PC, the synchronization takes place via Bluetooth or USB connectivity.

A data modification or update at one node or server must be reflected at all other (or target) nodes or servers. This is known as one-to-many synchronization.

Data copies between the server (one) and the nodes should be consistent or identical (many). Inconsistent copies of the same piece of information should not exist on the communicating node or server or on any other node or server.

A modification to data at one node or server must be replicated to all other nodes or servers (the target nodes or servers). Between several nodes, this is referred to as "many-to-many data synchronization." The data should remain consistent or identical across all nodes. There shouldn't be multiple copies of the same data on nodes or servers.

2.8 MOBILE COMPUTING AND THE APPS

The new standard, HTML5, allows for the design of mobile browsers and the creation of mobile applications on the Internet. It also offers additional functionalities.

Utilizing Mobile IP or WAP gateways of direct HTTP applications over 3G data services, users can access the internet for computing. A comprehensive suite called Mobile Enterprise Application Platform (MEAP) is utilized for mobile computing via the Internet. It makes it possible to create mobile applications, which need Web services, servers, and enterprise servers.

A mobile middleware server, application software, and a mobile client API make up a MEAP system. Uses for a middleware server include:

1. Administration of back-end servers' or systems' data
2. Setting up the app for the mobile device
3. System integration
4. Scalability

5. Communications
6. Support across platforms
7. Security

Mobile web applications and modern web applications: HTML5 is utilised for existing applications that are mobile-friendly and relate to web pages and web forms. Numerous capabilities and mobile-friendly controls are available with Microsoft ASP.Net 2.0. ASP.NET MVC tools are used to make the current Web applications mobile-friendly (MVC stands for model, view, and control).

Mobile makes it possible to employ cross-platform development tools, automate testing for tablets, iOS, Android, and Windows 8 devices, and integrate with existing applications as necessary. Analytics can also be used to enhance current apps. (Data analytics refers to the application of methods for drawing inferences following the examination of raw data and the conclusion of the information gathered.)

2.9 NOVEL APPLICATIONS OF MOBILE COMPUTING

There are numerous uses for mobile computer devices. Mobile TV is now a reality because to mobile computing. iPad, tablet, and PCs with extremely high mobility. The size of a paperback book is now the standard for 6" laptops, e-books, and readers.

1. Smartphones

A Smartphone is a mobile phone with added processing capabilities that allow for several applications to be used. For instance, Research in Motion, Inc.'s BlackBerry 8530 curve includes additional computing capabilities that make it possible to use the following applications:

1. Phone, email, address book, MMS (multimedia messaging service), and SMS (short message service). Web surfing, a calendar, a to-do list, and a memo pad.
2. Support for well-liked Personal Information Management (PIM) applications
3. Viewing integrated attachments
4. QWERTY-style keyboard with Sure Type technology (a computer keyboard has keys in order of Q,W,E,R,T,Y,...)
5. Keyboard controls for Send and End
6. Use a headset, vehicle buds, or car kits with Bluetooth to talk hands-free.
7. When EV-DO compatibility is enabled, the device can be used as a wireless modem for a laptop or personal computer.
8. Speaker telephone

9. Polyphonic ringtones allow you to customize your gadget.
10. A vivid, high-resolution display with over 65,000 color options
11. E-Mail
12. 802.11b and 802.11g WiFi
13. Browser
14. GPS tracking
15. Recording and communication through media, including audio, video, and camera images
16. Live TV
17. Micro SD cards

2. SmartWatch and iWatch

The development of mobile computing, NFC, medical device design, smart sensor design, and GPS technology led to the creation of the Galaxy Gear S, an innovative application product that Samsung released in August 2014. It is a brand-new smartwatch. It is a brand-new smartwatch. These are some of its characteristics: It sports a curved 2-inch display. It offers the capacity to make phone calls without using a smartphone at all. It has Bluetooth and Wi-Fi connectivity options. It permits GPS. With its heart rate and UV sensors, it features the S Health app, which alerts the wearer when it's time to eat, when they've had enough activity, and when it's time to relax. It contains walking navigational features. It allows watch owners to send a text message from their wrist. Apple Watch provides apps for those who want to lead an active lifestyle, such as Nike+ Running, which allows users to log morning or evening runs.

3. Music, Video, and e-Books

Apps that let you download music, film, and books are available for tablets and other mobile devices. Reading one's favorite books whenever and wherever one wants is now accessible thanks to tablets.

AAC-LC plays stereo audio in.m4v,.mp3, and .mov file formats at up to 160 kbps and 48 kHz. Apps for tablets and other devices provide mobile TV and MPEG-4 video. E-books can be read as PDFs or PNGs.

4. Mobile Cheque and Mobile Wallet

A mobile-based payment method used when making a purchase is called a "mobile cheque" (m-cheque). The customer, a certain retail location, and the mobile service provider all text messages to activate the service. In order to transfer money to the retailer account, the service provider authenticates the consumer and activates the customer account. The method that payments are made for purchases is changing as a result of these mobile devices. Customers are no longer required to have credit cards in their wallets while they shop.

A mobile wallet is an application where you may access your money via your service provider's account. A bank is used to link the payment when a mobile device connects to the service. Both the client's and the recipient's bank accounts are connected to the provider.

5. Mobile Commerce

The following is an illustration of mobile commerce (m-commerce). Stock quotes can be accessed on demand or in real time using mobile devices. The stock purchaser or seller first sends an SMS for the trading request, then the stock trading service responds in the same manner, requesting authentication. The client delivers the user ID and password through SMS. The customer is then instructed to continue by way of a confirmation SMS. The client sends an SMS to request a certain stock trade. At the stock exchange terminal, the service provider places the order. Within a couple of minutes, the procedure is finished online.

A buyer communicating their intention to purchase a product to the provider of mobile purchase services via SMS is another example of m-commerce. The service provider sends the product's prices in increasing order of the price at various retailers that sell the same item. Following that, the client asks the service provider to send the order to the least expensive and closest supplier. Additionally, the usage of mobile devices for e-ticketing, or the purchase of movie, train, aero-plane, and bus tickets, is growing.

6. Mobile-based Supply Chain Management

Apps for managing the supply chain on mobile devices effectively manage manufacturing, supply, logistics, warehouse, and other tasks while on the go. The following example provides the clearest explanation of the supply chain management issue. Chocolates cannot be sold by distributors unless they are manufactured. Without a distributor order, the manufacturer cannot produce the chocolates. The supply chain management problem is the name given to this producer-consumer issue. Leading IT firms have created supply chain management system software for mobile devices. Such mobile devices are used by the sales force and the manufacturing facilities to maintain the supply chain.

EXERCISE:

Question 1: Write about the features in a handheld computer.

Question 2: What purposes does data dissemination serve? Why is data distribution required to keep a mobile service running?

Question 3: Describe data synchronization

Question 4: Write short notes on the various cryptographic algorithms.

Question 5: What are the different limitations of using mobile devices?

2.10 SUMMARY

An access point may serve as a bridge connecting wired and wireless networks. It is a system that connects mobile systems and embedded systems to wireless LAN, the Internet, or the network of a mobile service provider.

Base station refers to a transceiver that connects wirelessly to a number of mobile devices or access points and wired or fibre optic connections to mobile switching centres and other networks.

With data rates of up to 1 Mbps, Bluetooth is a standard for object exchange that enables short-range (1 m or 100 m depending on radio spectrum) mobile communication between wireless electronic devices (for instance, between a mobile phone handset and headset for hands-free talking, for connecting the computer or printer, etc.).

By using multiplexing and other approaches, the capacity of the frequency channels—the number of users who can be supported at a particular time—is increased.

A computational structure where a client asks for computations and data, and after the necessary computations, the client receives the needed data or replies It is a computational structure in which, following computations at a server, the client caches or reads the data record(s). Access may occur upon client request, via broadcasts from the server, or via distribution. The client and server may be running on the same computer system or may be running on distinct computers.

Computation carried out on a mobile device while running an application in which a number of servers from a service provider or distributed computing systems take part, connect, and synchronise using mobile communication protocols.

By adopting formats that allow for playback on mobile devices or PCs, multimedia files (such as those for audio programmes or music videos) can be distributed via the Internet using the podcasting technique.

A protocol is a generally accepted recommended procedure for managing and regulating the transmission of data as well as the rules governing the syntax, semantics, and synchronisation of communication between two computing systems. Protocols also establish connections, format and sequence data, address the destination and sources, and terminate connections.

The integration of computations with environment items that have computing capabilities is referred to as ubiquitous computing.

WLAN is a wireless LAN that communicates using sets of common protocols, including IEEE 802.11a, 802.11b, and 802.11g.

Applications for mobile computer systems are numerous. Mobile TV has just lately become a reality thanks to mobile computers. A paperback book-sized version of the ultra-portable PC, tablet, iPad, 6" laptop, e-book, and reader is now available.

2.11 FURTHER READINGS

https://www.techtarget.com/searchmobilecomputing/definition/WPAN

https://mjginfologs.com/application-of-mobile-computing/

"Mobile Computing" by Asoke K Telukder, Roopa R Yuvagal, TMH

FUNDAMENTALS OF MOBILE COMMUNICATION by Mehaboob Mujawar, Jafar A. Alzubi

Introduction to Mobile Communication

S Sureshkumar, Fr. J. Janet, APS. Anandaraj

Wireless And Mobile Communication by Sanjeev Kumar, New Age International (P) Ltd., Publishers

MOBILE AND WIRELESS COMMUNICATION

By Leena R. Mehta

Mobile Communication

by: Behera G. K.